

While many measures of centrality have been proposed, the category itself is not well defined beyond general descriptors such as node **prominence** or structural **importance**. In addition, people propose all kinds of interpretations of centrality measures, such as (potential for) **autonomy, control, risk, exposure, influence, belongingness, brokerage, independence, power** and so on. The one thing that all agree on is that centrality is a node-level construct. But what specifically defines the category? What do all centrality measures have in common? Are there any structural properties of nodes that are not measures of centrality? (Borgatti and Everett 2006: 467).

Centrality
Measure
Decision
Tree

**Radial
Measures**

Volume Measures

K-path (degree) centrality
Eigenvector centrality
Friedkin's Immediate Effects Centrality

Edge-disjoint
(Walk Type (3))
Vertex-disjoint

(Walk Property (2))

Length Measures

Freeman's Closeness Centrality
Centroid Centrality
Friedkin's Total Effects Centrality
Euclidian (metric distance) Centrality

(Walk Position (1))

**Medial
Measures**

Volume Measures

Freeman's Betweenness Centrality
Newman's Random Walk Centrality
K-path Betweenness
Flow Betweenness

(Walk Property (2))

Length Measures

Borgatti's Distance-
Weighted Fragmentation
Centrality

Geodesic
(Walk Type (3))
All paths

Adapted from: Borgatti and Everett, 2006, *Social Networks*

How to compute a radial centrality measure

- Use the original adjacency matrix **A** to construct a matrix **W** that summarizes the strength of dyadic interactions across all pairs nodes.
 - $w_{ij}=1/d_{ij}$, where d_{ij} is the distance between node i and node j .
 - The distance measure is determined by the *walk type* and the *walk property* chosen:
 - Number of geodesics
 - Number of k-paths
 - Weighted measures
- Decide on an aggregation strategy:
 - Sum across rows (e.g. $c_j=\sum w_{ij}$).
 - Row average (e.g. $c_j=1/k_i\sum w_{ij}$).
 - Row median, mode, maximum, etc.

How to compute a medial centrality measure

- Use the original adjacency matrix $\mathbf{A}$ to construct a matrix $\mathbf{W}$ that summarizes the strength of dyadic interactions across all pairs nodes.
 - w_{ij} is the total number of paths through which something can flow or be transmitted from node i to node j .
- Construct a new matrix ${}^k\mathbf{W}_{i,j}$, with row k and column k removed (with k being the node that we are interested in computing the centrality measure for).
- Construct a new matrix ${}^k\mathbf{W}^*_{i,j}$ that summarizes the strength of dyadic interactions across all pairs nodes without including the focal node.
- Compute the quantity: $c_k = 1/{}^k\mathbf{W}_{ij}(\sum {}^k\mathbf{W}_{ij} - {}^k\mathbf{W}^*_{ij})$
- Repeat for each node k .

Walk Position				
Radial			Medial	
Walk Property			Walk Property	
<i>Volume</i>	<i>Length</i>		<i>Volume</i>	<i>Length</i>
Prominence	Exposure (Indirect)		Autonomy	Control
Influence (Direct)	Influence (Direct and Indirect)		Influence (Indirect)	Influence (Indirect)
Structural Importance	Belongingness (Indirect Integration)		Risk (Direct)	Risk (Indirect)
Exposure (Direct)			Structural Importance (Direct)	Structural Importance (Indirect)
Integration (Direct)			Independence	Brokerage

Note: The “**what**” and the “**how**” of the flow through the walk structure determines the *walk type* that you choose (Borgatti 2005).